Empirical Methods (in Finance)

Lecture 1 - Random Walks, Market Efficiency & Event Studies

Prof. Dr. Gianluca De Nard | Prof. Dr. Sebastian Stöckl

2026-02-01

Table of contents

1 Random Walks **1**

2 Testing for Random Walks **12**

3 Market Efficiency and Empirical Findings **22**

4 Event Studies **27**

1 Random Walks

1.1 Learning Outcomes

- Understand the concept of random walks and martingales in financial markets.
- Distinguish between different forms of market efficiency (weak, semi-strong, strong).
- Explain the link between random walks, martingales and the Efficient Market Hypothesis (EMH).
- Set up and estimate simple tests of the random walk hypothesis in R.
- Understand the basic structure of an event study and the notion of abnormal returns.
- Implement a simple event study in R and interpret cumulative abnormal returns (CARs).

By the end of this lecture, you should be able to clearly define random walks and martingales, relate them to the EMH, and critically discuss what it means for prices to follow a random walk in practice. You should also have a first hands-on experience in R with simulating and testing random walks and with implementing the basic steps of an event study (estimating a market model, computing abnormal and cumulative abnormal returns, and evaluating their statistical significance).

1.2 Introduction to Market Efficiency & Random Walks

- **Concept:** Random Walks in financial markets
- **Historical development:** From Bachelier to modern Geometric Brownian Motion (GBM)
- **Empirical implications:** Testing market efficiency using statistical methods

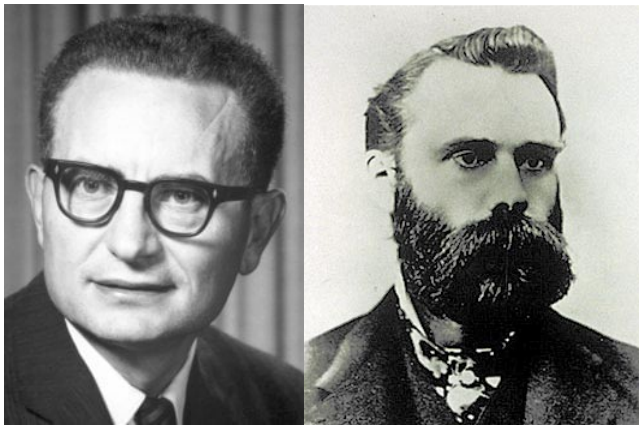
Key points: **Random Walks**, **Market Efficiency** and **Event Studies**

In today's lecture, we delve into the foundations of modern empirical finance by exploring the concept of market efficiency and the random walk hypothesis. We begin with the historical development from Bachelier's pioneering work in 1900 (Bachelier, 1900) to Samuelson's influential contributions in 1965 (Samuelson, 1965), and later to the development of the Black-Scholes model in 1973 (Black & Scholes, 1973). This historical progression has shaped the way we understand asset price dynamics and the role of randomness in financial markets.
Key points include:

- **Random Walks:** This hypothesis suggests that asset prices evolve in an unpredictable manner, akin to a random walk.
- **Market Efficiency:** According to Fama (Fama, 1970), if markets are efficient, then asset prices fully reflect all available information, making it impossible to consistently earn excess returns through arbitrage.
- **Empirical Tests:** We will later explore how statistical tests (e.g., variance ratio tests, unit root tests) are used to verify these concepts.

This lecture will serve as the basis for understanding more advanced topics later in the course.

1.3 History of Geometric Brownian Motion (GBM)



- Bachelier (1900)**:** Introduced the concept of Brownian motion for **modeling stock prices**.
- **Martingales:** Evolved as a critical concept in probability theory and finance.
- Samuelson (1965)**:** Adapted these ideas, promoting the notion that price changes should be **unpredictable**.
- Black & Scholes (1973)**:** Established GBM as the backbone for option pricing models.

Louis Bachelier's seminal work in 1900 (Bachelier, 1900) introduced the idea that stock prices could be modeled using Brownian motion—a continuous-time stochastic process. Although his work initially went largely unnoticed, it laid the groundwork for later developments.
Paul Samuelson in 1965 (Samuelson, 1965) took these ideas further by proposing that asset price movements could be best described by a martingale process. A martingale is a process where the best estimate for tomorrow's value is today's value, implying that past information cannot be used to predict future movements—a cornerstone in the efficient market hypothesis (EMH).
The Black-Scholes model (Black & Scholes, 1973) builds on GBM by incorporating both drift and volatility, making it a central model in pricing derivatives. In our lecture, we will discuss these models in detail, including the assumptions behind them and their empirical implications.

1.4 Martingales and the Martingale Hypothesis

A stochastic process $\{P_t\}$ is a **martingale** if it satisfies:

$$E[P_{t+1} \mid P_t, P_{t-1}, \dots] = P_t.$$

Or equivalently:

$$E[P_{t+1} - P_t \mid P_t, P_{t-1}, \dots] = 0.$$

- **Interpretation:** Today's price P_t is the best predictor of tomorrow's price.
- **Implication for Markets:** If asset prices are martingales, then no strategy based on past information can consistently outperform the market.

The martingale property is central to our understanding of market efficiency. Mathematically, the above definition means that the expected incremental change in the price, given all past information, is zero.
Key Symbols Explained:

- P_t: Price of the asset at time t.
- $E[\cdot]$: The expectation operator.
- The condition $E[P_{t+1} \mid P_t, P_{t-1}, \dots]$ implies that the forecast of P_{t+1} using all available information is exactly P_t.

This means that if the market follows a martingale, then any deviation from P_t is entirely random and not exploitable. This is one of the core ideas behind the Efficient Market Hypothesis as outlined in Fama (1970) (Fama, 1970).

1.5 Implications of the Martingale Hypothesis

- **No Arbitrage:** Prices P_t quickly adjust to new information, precluding riskless profit.
- **Full Reflection of Information:** All known information is already embedded in current prices.
- **Limited Predictability:** Past prices $P_{t-1}, P_{t-2}, \dots$ cannot be used to predict future returns reliably.

When asset prices follow a martingale process, it implies that any attempts at predicting future price movements based on past data are futile. This is because all past information is already reflected in the current price P_t. Consequently, arbitrage opportunities—situations where a riskless profit can be made—are eliminated.
This property underpins the Efficient Market Hypothesis (EMH) (Fama, 1970), which has been a subject of extensive empirical testing and debate. Our exploration today will further elaborate on how these theoretical constructs can be empirically tested.

1.6 From Martingales to Random Walks

- **Martingales:** Allow for randomness but do not require the noise term ϵ_t to be independently distributed.
- **Random Walks:** A stricter assumption where price increments are independent and identically distributed (IID):
$$P_t = P_{t-1} + \epsilon_t,$$
where:
  - ϵ_t is an IID random variable, typically assumed to follow a normal distribution:
$$\epsilon_t \sim N(0, \sigma^2).$$

In the random walk model, the key assumption is that the error terms ϵ_t are independent and identically distributed (IID). This implies that each ϵ_t is drawn from a normal distribution with mean 0 and variance σ^2.
Explanation of Symbols:

- P_t: The asset price at time t.
- ϵ_t: The stochastic error term, representing the unpredictable component of price changes.
- $N(0, \sigma^2)$: Denotes a normal distribution with mean 0 and variance σ^2.

This framework suggests that today's price change provides no information about tomorrow's price change, thereby reinforcing the idea that asset returns are unpredictable in an efficient market.

1.7 Random Walk: Motivation & Intuition

- **Short-run memory:** If prices follow a random walk, there is no "memory" of past movements.
- **Fair Game Interpretation:**
  - $E[P_{t+1} \mid P_t, ...] = P_t$
  - No systematic way to profit from past prices.
- **Distributional Assumptions:**
  - Often assume $\epsilon_t \sim N(0, \sigma^2)$
  - Implies P_t is non-stationary (variance grows with t).

A **random walk** can be thought of as a "drunkard's walk": each step is independent of the last. From a finance perspective, this implies **no short-run memory** in price changes and suggests that the best estimate for tomorrow's price is today's price. In practice, we often assume normally distributed increments, although empirical data sometimes exhibit fatter tails.

1.8 The Random Walk Hypothesis

- **Historical Origin:** The concept dates back to Bachelier (1900).
- **Key Assumption:** Future price changes are independent of past movements.
- **Mathematical Formulation:**

$$E[P_{t+1} \mid P_t, P_{t-1}, \dots] = P_t.$$

- **Empirical Implication:** Asset prices are inherently unpredictable.

The random walk hypothesis is a direct extension of the martingale property but assumes that the increments (ϵ_t) are IID. This means that:

- There is no memory in the process.
- Each price change is a fresh draw from the same probability distribution.

This hypothesis has been the subject of numerous empirical tests. Researchers have used various statistical techniques—such as variance ratio tests and autocorrelation tests—to validate or refute this hypothesis (see (Lo & MacKinlay, 1988) for one influential study). Our subsequent slides will discuss these methods in depth.

1.9 Arithmetic Brownian Motion

Arithmetic Brownian Motion (ABM):

$$P_t = P_{t-1} + \mu + \epsilon_t,$$

- μ: drift component (expected price change = **compensation for risk**).
- σ: Constant volatility.
- ϵ_t: $IID(0, \sigma^2)$ noise term.
  - **Independence** of ϵ_t implies that increments of different periods are uncorrelated (a fair game)
  - **Identically** implies that the distribution of increments does not change over time
- Through **recursive substitution** we show:
  $E[P_t|P_0] = P_0 + \mu t$ and $Var[P_t|P_0] = \sigma^2 t$

1.10 Geometric Brownian Motion

Violation of the **limited liability of equity**:

- If ϵ_t are normally distributed, P_t may become negative
- **Solution**: We use the natural logarithm of prices, where $p_t = \ln(P_t)$:

$$ln(P_t) = \mu + ln(P_{t-1}) + \epsilon_t$$

$$p_t = \mu + p_{t-1} + \epsilon_t$$

- This is Geometric Brownian Motion (GBM)
- Continuously compounded returns are normally distributed with μ and σ^2
- **Key Difference:** GBM models relative (percentage) changes, ensuring non-negative prices.

Arithmetic Brownian Motion is a linear model which, while simpler, can lead to negative prices—an unrealistic outcome in financial markets. In contrast, GBM models the logarithm of prices, which naturally prevents negative values.
GBM is used extensively in the Black-Scholes model (Black & Scholes, 1973) for pricing options because it realistically models how returns, rather than absolute price changes, behave.
The transition from a simple random walk to GBM introduces proportional changes in the asset price. In GBM, the drift μP_t implies that returns, not absolute changes, drive the price evolution. Likewise, the volatility term σP_t scales the noise according to the price level, ensuring that the model reflects real-world characteristics such as volatility clustering.

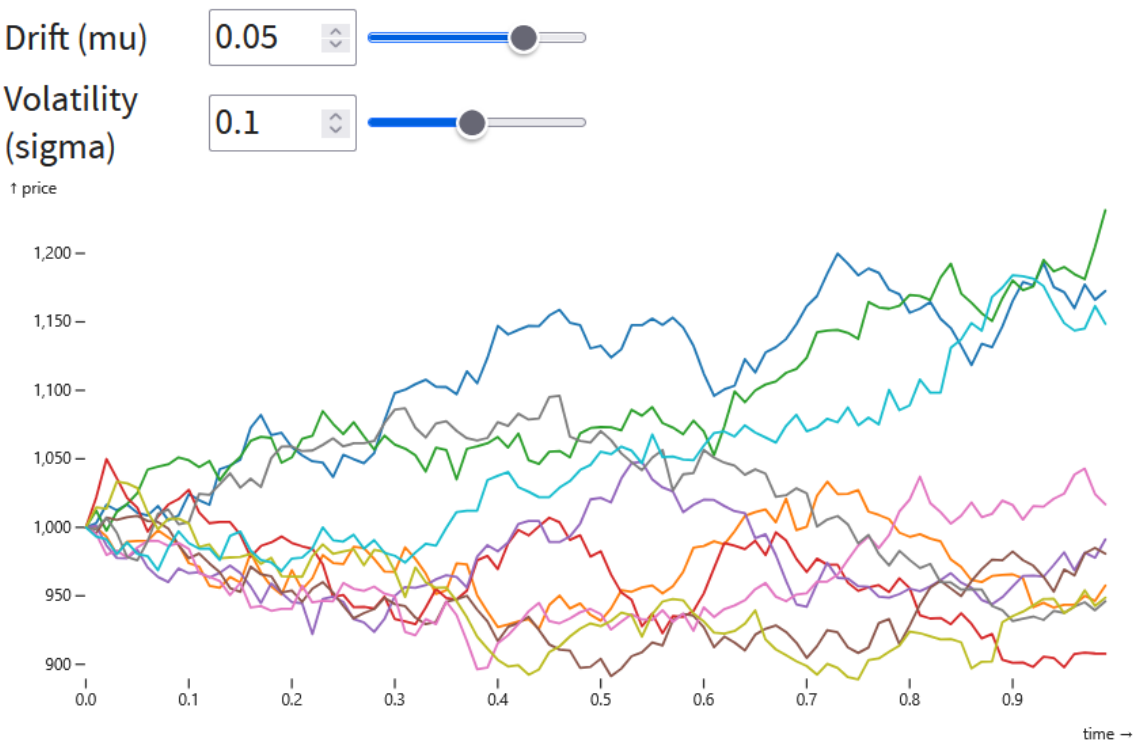
1.11 Properties and Implications of Random Walks

- **Non-stationarity:**
  If P_t follows a random walk,
  $$P_t \sim N(\mu t, \sigma^2 t).$$
- **Variance Growth:**
  Variance increases linearly over time:
  $$\mathrm{Var}(P_t) = t\sigma^2.$$
- **No Mean Reversion:** Prices do not tend to return to a long-term average.

One of the fundamental properties of a random walk is its non-stationarity. As time increases, the variance of the price distribution grows without bound, which means that the uncertainty about future prices increases linearly with time.
This property has practical implications for forecasting and risk management, as long-term predictions become increasingly uncertain. Additionally, the absence of mean reversion implies that once a trend has started, there is no inherent force that pulls prices back to an average level.

1.12 Interactive Simulation of 10 GBM Paths

Below, you can set drift (μ) and volatility (σ) to visualize 10 simulated GBM paths:



1.13 Three Versions of Market Random Walks

(Adapted from Lo & MacKinlay (1988))

1. **RW1:** Increments are *independent and identically distributed* (IID).

2. **RW2:** Increments are independent but *not* identically distributed.

3. **RW3:** Increments are *uncorrelated* but may exhibit higher-order dependencies.

Implications:

- **Stronger assumptions** (as in RW1) imply less predictability.
- Empirical evidence often rejects RW1 but may support RW3.

> Lo & MacKinlay (1988) provided a classification of random walks to better capture the complexity observed in real-world financial data. The strictest form (RW1) assumes complete IID behavior, while RW3 relaxes this assumption to allow for some forms of dependence (e.g., in higher moments), yet still maintains that there is no linear predictability.
> This nuanced view helps explain why some empirical tests might reject the pure random walk hypothesis while still showing uncorrelated returns.

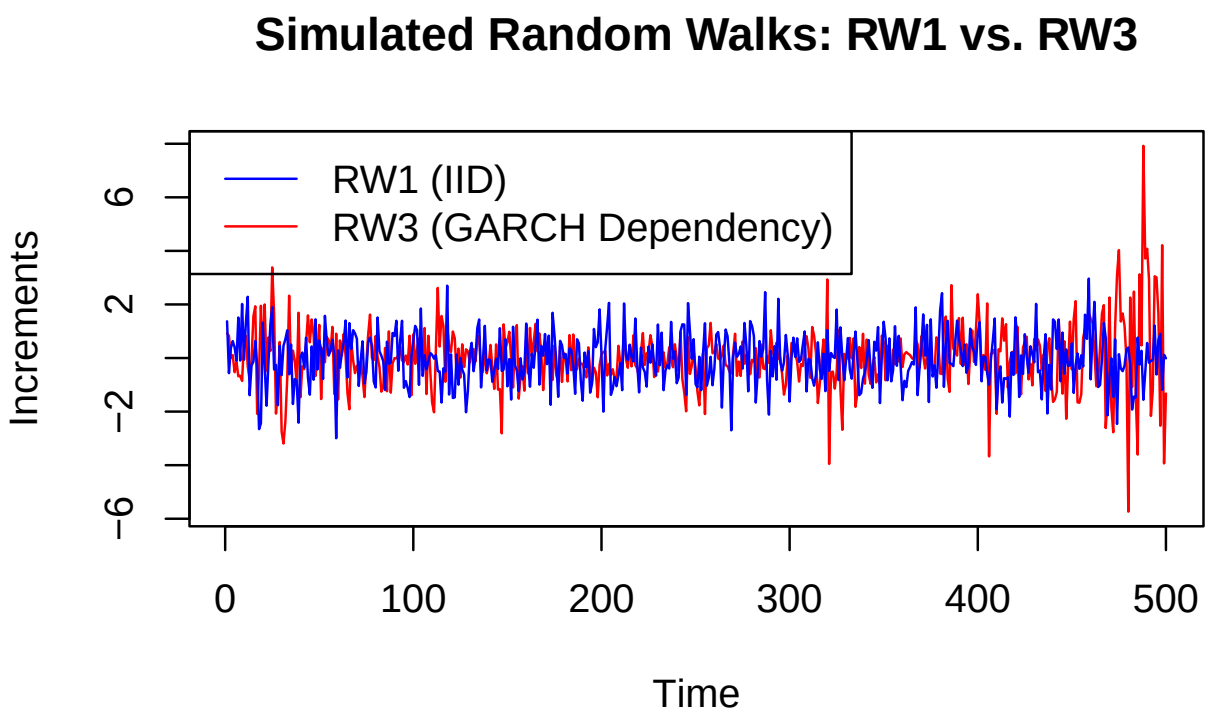
1.14 R Code: Simulating RW1 vs. RW3

```r
set.seed(42)
n.sim <- 500

# Simulate RW1: IID increments
rw1_inc <- rnorm(n.sim, mean = 0, sd = 1)
rw1 <- cumsum(rw1_inc)

# Simulate RW3: Introducing dependence via an GARCH(1,1) process
# install.packages("rugarch")  # Run this if you haven't installed it yet
library(rugarch)
```

```
Loading required package: parallel
```

```r
spec <- ugarchspec(
  variance.model = list(model = "sGARCH", garchOrder = c(1, 1)),
  mean.model = list(armaOrder = c(0, 0), include.mean = FALSE),
  fixed.pars = list(omega = 0.05, alpha1 = 0.20, beta1 = 0.75)
)
```

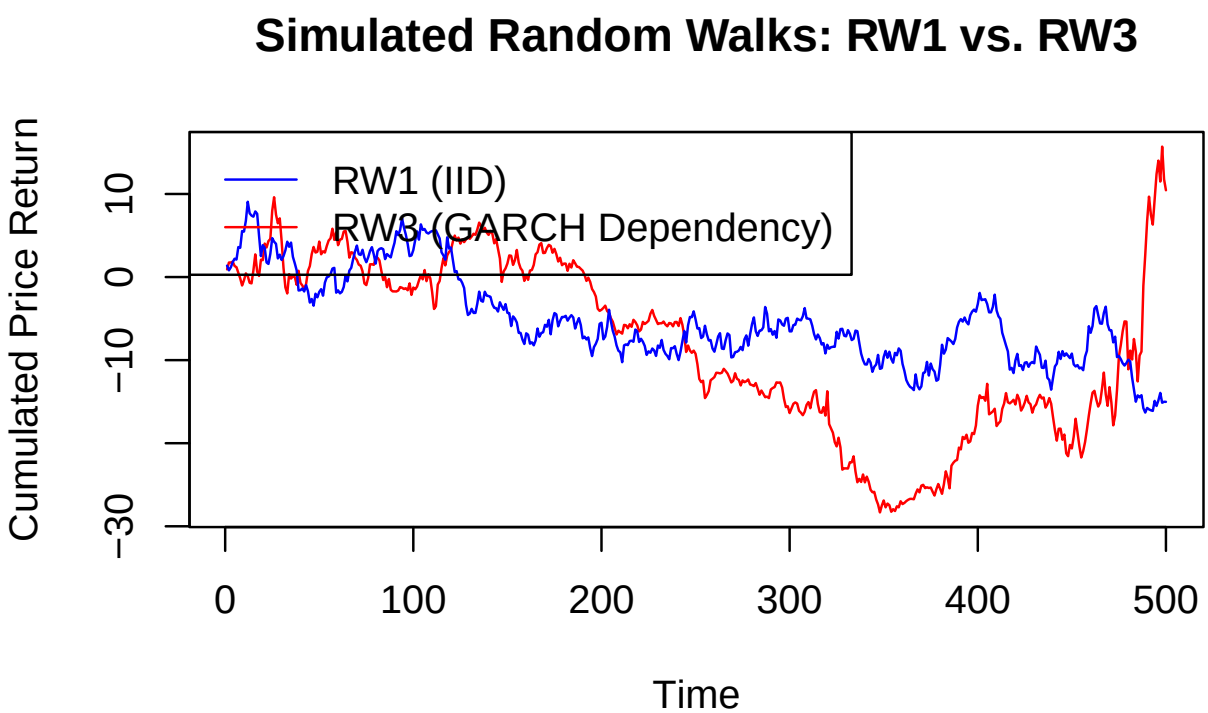
```r
sim <- ugarchpath(spec, n.sim = n.sim)
garch_series <- fitted(sim)

rw3 <- cumsum(garch_series)


plot(garch_series,
  type = "l", col = "red", main = "Simulated Random Walks: RW1 vs. RW3",
  ylab = "Increments", xlab = "Time"
)
lines(rw1_inc, col = "blue")
legend("topleft",
  legend = c("RW1 (IID)", "RW3 (GARCH Dependency)"),
  col = c("blue", "red"), lty = 1
)
```



1.15 R Code: Simulating RW1 vs. RW3

```r
plot(rw3,
  type = "l", col = "red", main = "Simulated Random Walks: RW1 vs. RW3",
```

```r
  ylab = "Cumulated Price Return", xlab = "Time"
)
lines(rw1, col = "blue")
legend("topleft",
  legend = c("RW1 (IID)", "RW3 (GARCH Dependency)"),
  col = c("blue", "red"), lty = 1
)
```



This simulation contrasts two types of random walks:

- **RW1:** Uses IID increments, where each increment is independent and drawn from $N(0, 1)$.
- **RW3:** Introduces some higher order dependeny by using a GARCH(1,1) process (second order dependency / volatility clustering.

By visualizing these processes, you can see how dependence in the increments alters the trajectory of prices. Experimenting with these parameters in R helps to solidify your understanding of these theoretical concepts.

2 Testing for Random Walks

2.1 Testing for Random Walks

- **Why test for random walks?**
  - Validates market efficiency in its simplest form
  - Determines if future prices can be predicted from past data
- **Recap of RW1, RW2, RW3** (Lo & MacKinlay, 1988)
  1. **RW1:** Strict IID increments
  2. **RW2:** Independent but not identically distributed
  3. **RW3:** Uncorrelated increments (no linear predictability)

Up to now, we have introduced the concept of a random walk and discussed how asset prices might or might not exhibit memory. In this section, we **bridge** that theoretical foundation to the **practical tests** used to detect violations of the random walk hypothesis. We'll see how each test can reveal different types of dependence or non-IID behavior in price data.

2.2 Which Test for Which RW Version?

- **Runs Test**
  - Checks the randomness of sign changes or above/below median sequences
  - Generally **catches** broad deviations from independence (RW1/RW2)
- **Autocorrelation Tests** (Ljung-Box, Box-Pierce)
  - Target **linear** dependencies in returns
  - Distinguish between **uncorrelated** vs. correlated increments (addresses RW3)
- **Variance Ratio Tests** (Lo & MacKinlay)
  - Specifically designed to detect if **variance scales linearly** with time

- Often used to test **strict** random walk assumptions (RW1, but also informative for RW2)

Each test is sensitive to different facets of the random walk assumptions:

1. **Runs test** is a **nonparametric** method that picks up general deviations from randomness. If the series is not even "random" in the sense of sign changes, it likely fails **RW1** or **RW2**.
2. **Autocorrelation tests** examine **linear** relationships. If returns are autocorrelated, that directly violates the uncorrelated increments assumption (particularly **RW3**).
3. **Variance ratio** focuses on how multi-period variance compares to single-period variance, a hallmark of **IID** increments. Failure indicates a potential breakdown of **RW1** or **RW2** assumptions.

2.3 Runs Test

- **Purpose:** Detect randomness in a sequence by counting "runs" of consecutive positives/negatives.
- **Hypothesis:**
  - H_0: Data is generated by a random process (no memory).
  - H_1: Sequence exhibits non-random patterns.
- **Interpretation:**
  - Too few runs $\rightarrow$ potential clustering
  - Too many runs $\rightarrow$ potential oscillation

The **runs test** helps pick up broad signs of dependence—be it in sign changes or above/below median transitions. If we can reject the null of randomness, the data might violate **RW1** or **RW2** (since independence is broken).

2.4 Autocorrelation Tests (1 of 2)

- **Goal:** Detect linear dependence over time lags

- **Null Hypothesis (H_0)**: No autocorrelation at each lag (i.e., returns are uncorrelated)
- **Common Tests:**
  - **Ljung-Box**: Incorporates a correction factor for small samples
  - **Box-Pierce**: Simpler, older variant

For a process to qualify as **RW3**, returns must be **uncorrelated**. Autocorrelation tests are directly aimed at uncovering any linear relationship across lags.

2.5 Autocorrelation Tests (2 of 2)

- **Ljung-Box Statistic:**

$$Q = n(n+2)\sum_{k=1}^{m} \frac{\hat{\rho}_k^2}{n-k}$$

  - $\hat{\rho}_k$: sample autocorrelation at lag k
  - Under $H_0 : Q \sim \chi_m^2$

- **Caveats:**
  - Choose lag m carefully
  - Large m can inflate Type I error

Rejecting the null means the increments (returns) have some autocorrelation, violating the "uncorrelated increments" assumption (especially **RW3**).

2.6 Variance Ratio Tests (1 v 2)

- **Idea:** Under a random walk, multi-period variance scales with time
- **Test Statistic:**

$$VR(q) = \frac{\mathrm{Var}(P_t - P_{t-q})}{q \cdot \mathrm{Var}(P_t - P_{t-1})}$$

  - If **RW** holds, $VR(q) \approx 1$ for all q

Lo & MacKinlay's variance ratio test is a powerful tool to check whether increments are truly IID. If the variance ratio differs significantly from 1, it suggests **RW1** (and possibly **RW2**) is violated.

2.7 Variance Ratio Tests (2 v 2)

- **Null Hypothesis** (H_0): $VR(q) = 1$ (random walk)
- **Distribution:**
  - Under H_0, $VR(q)$ has an asymptotic normal distribution (with certain corrections).
- **Interpretation:**
  - $VR(q) > 1$: Suggests positive serial correlation
  - $VR(q) < 1$: Suggests mean reversion or negative correlation

While the variance ratio test specifically addresses the time-scaling property of variance, keep in mind real markets may show volatility clustering or heteroskedasticity, which can complicate test results. Nevertheless, it remains a standard tool for random walk diagnostics.

2.8 Putting It All Together

- **Step 1:** Runs Test $\rightarrow$ Quick check of basic randomness
- **Step 2:** Autocorrelation Tests $\rightarrow$ Are returns linearly independent?
- **Step 3:** Variance Ratio $\rightarrow$ Does variance scale linearly with time?

In practice, researchers often apply **multiple tests** to see if the data pass all the conditions implied by **RW1**, **RW2**, or **RW3**. A market that fails the runs test may still have zero autocorrelation but exhibit other forms of dependency, etc. The synergy of these tests offers a comprehensive view.

2.9 Empirical Application (1 of 3)

```r
# Load the saved dataset
# Load required packages
library(tidyquant)
suppressPackageStartupMessages(library(tidyverse))
library(lubridate)
library(ggplot2)
library(randtests) # For runs.test
library(knitr) # For pretty tables

# Load the saved stocks data
load("stocks.RData")

symbol <- "AAPL"

# Filter for one stock (e.g., AAPL)
aapl <- stocks_data %>% filter(symbol == !!symbol)

# Define the frequencies to analyze
frequencies <- c("daily", "monthly", "quarterly", "yearly")

# Create an empty list to store results
results_list <- list()

for (freq in frequencies) {
  # Compute arithmetic returns for the chosen frequency
  ret <- aapl %>%
    tq_transmute(
      select = adjusted,
      mutate_fun = periodReturn,
      period = freq,
      col_rename = "returns"
    ) %>%
    drop_na()

  # Perform the runs test on returns
  runs_res <- runs.test(ret$returns)

  # Perform the Ljung-Box test (using min(10, n-1) lags)
  lag_n <- min(10, nrow(ret) - 1)
  box_res <- Box.test(ret$returns, lag = lag_n, type = "Ljung-Box")
```

```r
# For the variance ratio test we need different frequency returns
# in case freq is daily we calculate weekly returns and calculate
# the VR accordingly,
# in case of monthly we use weekly returns and
# in case of quarterly and yearly we relate to monthly
if (freq == "daily") {
  ret_alt <- aapl %>%
    tq_transmute(
      select = adjusted,
      mutate_fun = periodReturn,
      period = "weekly",
      col_rename = "returns"
    ) %>%
    drop_na()
  # Compute Variance Ratio for two different frequencies
  q <- 5
  n_p <- nrow(ret)
  VR <- ifelse(n_p > q, var(ret_alt$returns) / (q * var(ret$returns)), NA)
} else if (freq == "monthly") {
  ret_alt <- aapl %>%
    tq_transmute(
      select = adjusted,
      mutate_fun = periodReturn,
      period = "weekly",
      col_rename = "returns"
    ) %>%
    drop_na()
  ret2 <- ret
  # Compute Variance Ratio for two different frequencies
  q <- 4
  n_p <- nrow(ret)
  VR <- ifelse(n_p > q, var(ret$returns) / (q * var(ret_alt$returns)), NA)
} else if (freq == "quarterly" | freq == "yearly") {
  ret_alt <- aapl %>%
    tq_transmute(
      select = adjusted,
      mutate_fun = periodReturn,
      period = "monthly",
      col_rename = "returns"
    ) %>%
    drop_na()
  # Overlapping returns can be generated like this (for the annual returns)
```

```r
    if (freq == "yearly") {
      ret <- ret_alt |>
        tq_transmute(
          select = returns,
          mutate_fun = rollapply,
          width = 12,
          FUN = sum,
          col_rename = "returns"
        ) |>
        drop_na()
    }
    #
    q <- ifelse(freq == "quarterly", 3, 12)
    n_p <- nrow(ret)
    VR <- ifelse(n_p > q, var(ret$returns) / (q * var(ret_alt$returns)), NA)
  }
  # Save the results for this frequency
  results_list[[freq]] <- list(
    returns = ret,
    runs_test = runs_res,
    ljung_box = box_res,
    VR = VR
  )
}

# Create a summary table with key test results for each frequency
summary_table <- data.frame(
  Frequency      = frequencies,
  Runs_p_value   = sapply(results_list, function(x) x$runs_test$p.value),
  Box_p_value    = sapply(results_list, function(x) x$ljung_box$p.value),
  Variance_Ratio = sapply(results_list, function(x) x$VR)
)

# Display the summary table
print(kable(summary_table,
  digits = 4,
  caption = paste0("Summary of Random Walk Tests for ", symbol)
))

# Interpretative notes:
cat("\nInterpretation:\n")
cat("• A high p-value for the runs test suggests that the sequence
```

```
    of returns is random in terms of sign changes.\n")
cat("• A high p-value for the Ljung-Box test indicates that there
    is no significant linear autocorrelation.\n")
cat("• A Variance Ratio (VR) close to 1 supports the random walk
    hypothesis regarding the time-scaling property of log-prices.\n")
```

```
Table: Summary of Random Walk Tests for AAPL

|          |Frequency | Runs_p_value| Box_p_value| Variance_Ratio|
|:---------|:---------|------------:|-----------:|--------------:|
|daily     |daily     |       0.1178|      0.0000|         0.8650|
|monthly   |monthly   |       0.5630|      0.0769|         1.1870|
|quarterly |quarterly |       0.1594|      0.9506|         1.1880|
|yearly    |yearly    |       0.0950|      0.7495|         0.9935|

Interpretation:
• A high p-value for the runs test suggests that the sequence
    of returns is random in terms of sign changes.
• A high p-value for the Ljung-Box test indicates that there
    is no significant linear autocorrelation.
• A Variance Ratio (VR) close to 1 supports the random walk
    hypothesis regarding the time-scaling property of log-prices.
```

2.10 Empirical Application (2 of 3)

```
library(ggfortify)
```

```
Registered S3 method overwritten by 'ggfortify':
  method          from
  autoplot.glmnet parsnip
```

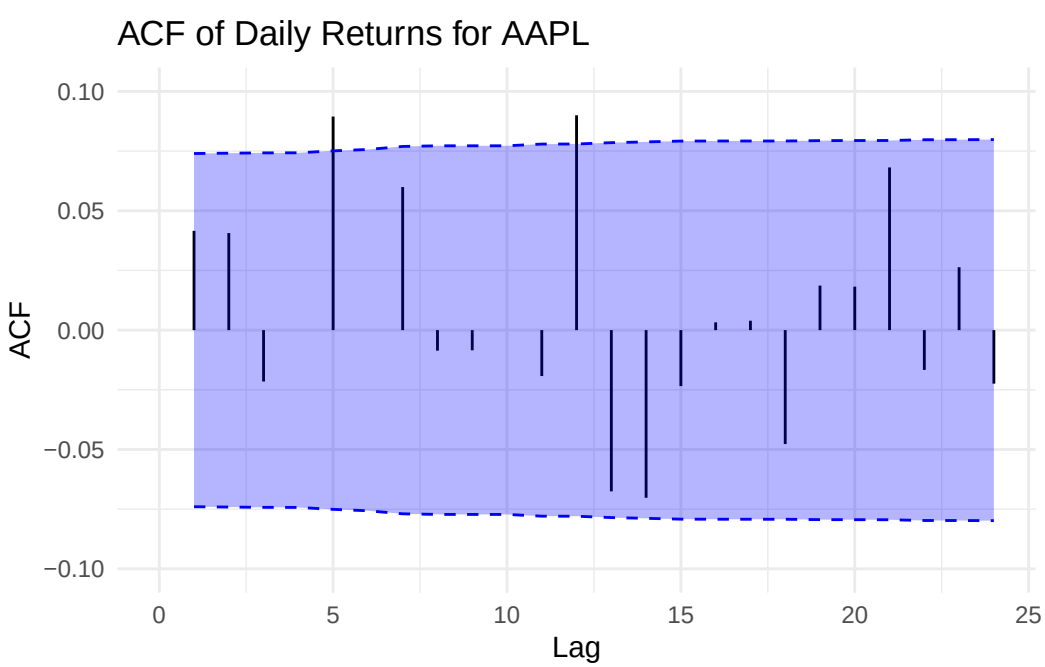
```
# Example: Plot the ACF for daily returns
daily_ret <- results_list[["monthly"]]$returns
acf_plot <- autoplot(acf(daily_ret$returns, plot = FALSE),
  conf.int.fill = "#0000FF", conf.int.value = 0.8,
  conf.int.type = "ma"
) +
  ggtitle(paste0("ACF of Daily Returns for ", symbol)) +
```

```
  ylim(c(-0.1, 0.1)) +
  theme_minimal()
```

```
Warning: `aes_string()` was deprecated in ggplot2 3.0.0.
i Please use tidy evaluation idioms with `aes()`.
i See also `vignette("ggplot2-in-packages")` for more information.
i The deprecated feature was likely used in the ggfortify package.
  Please report the issue at <https://github.com/sinhrks/ggfortify/issues>.
```

```
print(acf_plot)
```

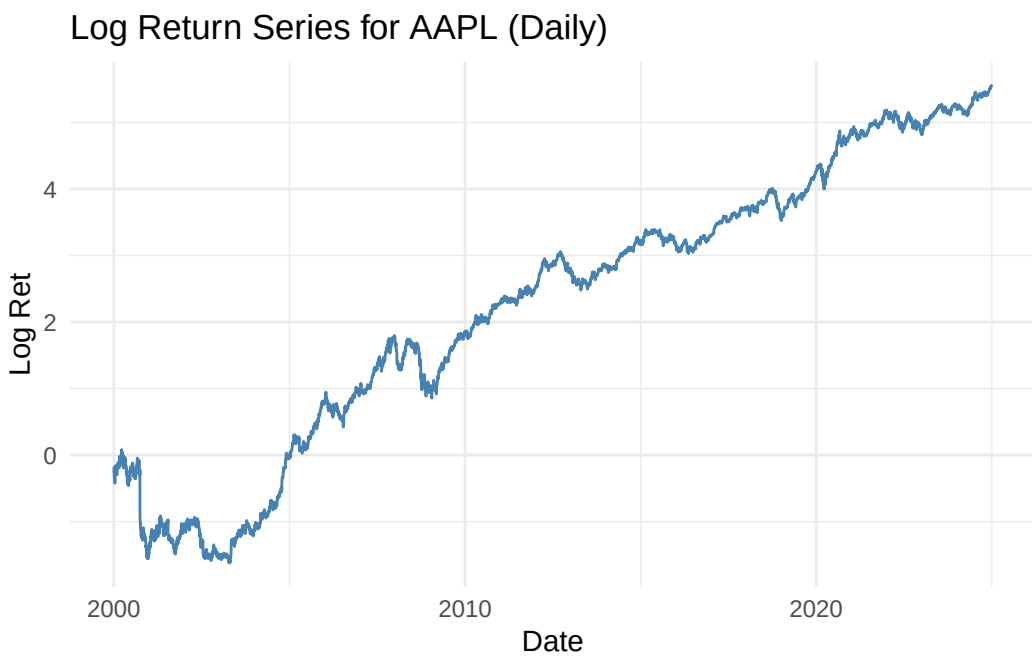
```
Warning: Removed 4 rows containing missing values or values outside the scale range
(`geom_segment()`).
```



2.11 Empirical Application (3 of 3)

```
# Plot the log-price series for daily frequency (for variance ratio context)
daily_log <- aapl %>%
  tq_transmute(
```

```r
    select = adjusted,
    mutate_fun = periodReturn,
    period = "daily",
    type = "log",
    col_rename = "log_return"
  ) %>%
  drop_na() %>%
  mutate(log_price = cumsum(log_return) + log(first(aapl$adjusted)))
logprice_plot <- ggplot(daily_log, aes(x = date, y = log_price)) +
  geom_line(color = "steelblue") +
  labs(
    title = paste0("Log Return Series for ", symbol, " (Daily)"),
    x = "Date", y = "Log Ret"
  ) +
  theme_minimal()
print(logprice_plot)
```



3 Market Efficiency and Empirical Findings

3.1 Definitions of Market Efficiency

- **Market Efficiency:**
  A market is efficient if asset prices reflect all available information.
- **Key Implication:**
  Investors cannot consistently earn abnormal returns by exploiting available information.

Market efficiency means that at any given time, prices reflect all relevant information. This idea implies that no investor can consistently outperform the market by trading on that information. The concept underpins modern finance theory.

3.2 History of Market Efficiency

- **Early Observations:**
  - Roberts (1967)**:** Early evidence on serial correlations suggested weak predictability in stock prices.
- **Formalization:**
  - Fama (1970)**:** Introduced the Efficient Market Hypothesis (EMH) with three forms.
- **Popularization:**
  - Malkiel (1989)**:** Advocated for passive index investing; "Random Walk" theory in mainstream finance.

Roberts' empirical studies in the 1960s laid the groundwork by questioning whether historical data could predict future returns. Fama (1970) later formalized these ideas, categorizing market efficiency into weak, semi-strong, and strong forms. Malkiel (1989) popularized these concepts and argued that actively managed funds rarely outperform the market.

3.3 Types of Market Efficiency

- **Weak Form Efficiency:**
  - Prices reflect all historical trading data (prices, volumes).
- **Semi-Strong Form Efficiency:**

  - Prices reflect all publicly available information (including financial statements, news).

- **Strong Form Efficiency:**
  - Prices reflect all information, public and private (including insider information).

The three forms of market efficiency differ in the information set that is assumed to be fully incorporated into prices. Weak form deals solely with past price and volume data; semi-strong extends to all public information; and strong form claims that even insider information is reflected in market prices.

3.4 What Market Efficiency Means

- **Implications of Efficiency:**
  - **No Arbitrage Opportunities:** No "free lunch" exists.
  - **Random Walk Behavior:** Future prices are unpredictable based on historical data.
  - **Fair Game:** Investors can only earn returns commensurate with the risk they take.

If markets are efficient, then the price of an asset is the best estimate of its true value given the information available. As a result, trading on historical or public information should not yield systematic excess profits.

3.5 What Market Efficiency Does Not Mean

- **It Does Not Mean:**
  - Prices are smooth or stable.
  - Investors never lose money.
  - All anomalies are opportunities for profit.

- **Clarification:**
  - Even in efficient markets, prices can be volatile due to new information and risk factors.

A common misconception is that market efficiency implies perfect predictability or constant upward price movement. In reality, efficiency means that information is quickly incorporated into prices—not that prices will be devoid of volatility or unpredictable movements.

3.6 The Information Paradox

Grossman & Stiglitz (1980):

- **Core Concept:**
  - If markets were perfectly efficient, there would be no incentive to gather costly information.
- **Implications:**
  - Some inefficiency must exist to reward information acquisition.
- **Empirical Impact:**
  - Explains why anomalies may persist, as full efficiency is not attainable.

Grossman & Stiglitz (1980) argued that if information were free and perfectly reflected in prices, no one would spend resources to obtain it. Hence, markets must be somewhat inefficient to compensate those who acquire information. This paradox explains why we sometimes observe persistent anomalies.

3.7 Testing Weak Form Efficiency

- **Time Series Tests:**
  - **Runs Test:** Nonparametric test of randomness.
  - **Autocorrelation Tests:** (Ljung-Box, Box-Pierce) Check for serial correlation.
  - **Variance Ratio Tests:** Examine scaling properties of variance.
- **Evidence:**
  - Empirical studies often find weak-form efficiency in developed markets.

Weak form tests examine whether historical price data can be used to predict future prices. In efficient markets, no test should reveal predictable patterns. We have seen these tests in previous slides.

3.8 Testing Semi-Strong Form Efficiency

- **Event Studies:**
  - Analyze market reaction to public news (e.g., earnings announcements, M&A events).
- Key focus: Speed and accuracy of price adjustments.
- **Portfolio Studies:**
  - Sort stocks by public information (e.g., P/E, book-to-market) and test for abnormal returns.
- **Interpretation:**
  - Quick and unbiased price adjustments support semi-strong efficiency.

Event studies measure how fast and accurately prices respond to new public information. If the market is semi-strong efficient, abnormal returns should be short-lived or non-existent after the event.

3.9 Testing Strong Form Efficiency

- **Insider Trading Studies:**
  - Compare returns earned by insiders versus non-insiders.
- **Disclosure Analysis:**
  - Assess whether private information is already reflected in prices.
- **Empirical Findings:**
  - Evidence generally indicates that insiders can earn abnormal returns, suggesting strong form efficiency does not hold.

Strong form efficiency is the most stringent version. Empirical evidence, particularly from insider trading studies, shows that insiders tend to earn abnormal returns, implying that not all private information is incorporated into market prices.

3.10 The Joint Hypothesis Problem

- **Definition:**
  - Tests of market efficiency are joint tests of both the efficiency hypothesis and the underlying asset pricing model.
- **Implications:**
  - Rejection of efficiency could be due to a flawed pricing model (e.g., CAPM) rather than true inefficiency.
- **Practical Considerations:**
  - Multiple models are often tested to disentangle model misspecification from market inefficiency.

The joint hypothesis problem is a central challenge in empirical finance. Since any test of market efficiency requires a model to determine the "fair" price, a failure of the test might reflect deficiencies in that model rather than a failure of the market to incorporate information.

3.11 Modern Evidence on Market Efficiency

- **Empirical Findings:**
  - **Weak Form:** Generally supported in liquid, developed markets.
  - **Semi-Strong Form:** Event studies typically show rapid incorporation of public information, although anomalies (e.g., post-earnings announcement drift) persist.
  - **Strong Form:** Evidence against strong efficiency is robust; insiders can earn abnormal returns.
- **Key References:**
  - Fama (1970), Malkiel (1989), and later studies continue to refine our understanding.
- **Ongoing Debate:**
  - Research continues to explore why anomalies persist (e.g., limits to arbitrage, behavioral biases).

Modern evidence suggests that while weak and semi-strong forms of market efficiency are largely supported, the strong form does not hold. Additionally, ongoing research highlights that market anomalies may be due to limits to arbitrage or behavioral factors rather than true inefficiency.

3.12 Final Takeaways

- **Market Efficiency is Multifaceted:**
  It ranges from weak to strong forms depending on the information set considered.
- **Empirical Tests Provide Mixed Evidence:**
  While many studies support the weak and semi-strong forms, the joint hypothesis problem and the information paradox complicate the interpretation.
- **Practical Implications:**
  For practitioners, these findings imply that while historical data may not reliably predict returns, some anomalies exist—but they might be too small to exploit once transaction costs and risks are accounted for.

4 Event Studies

4.1 Motivation

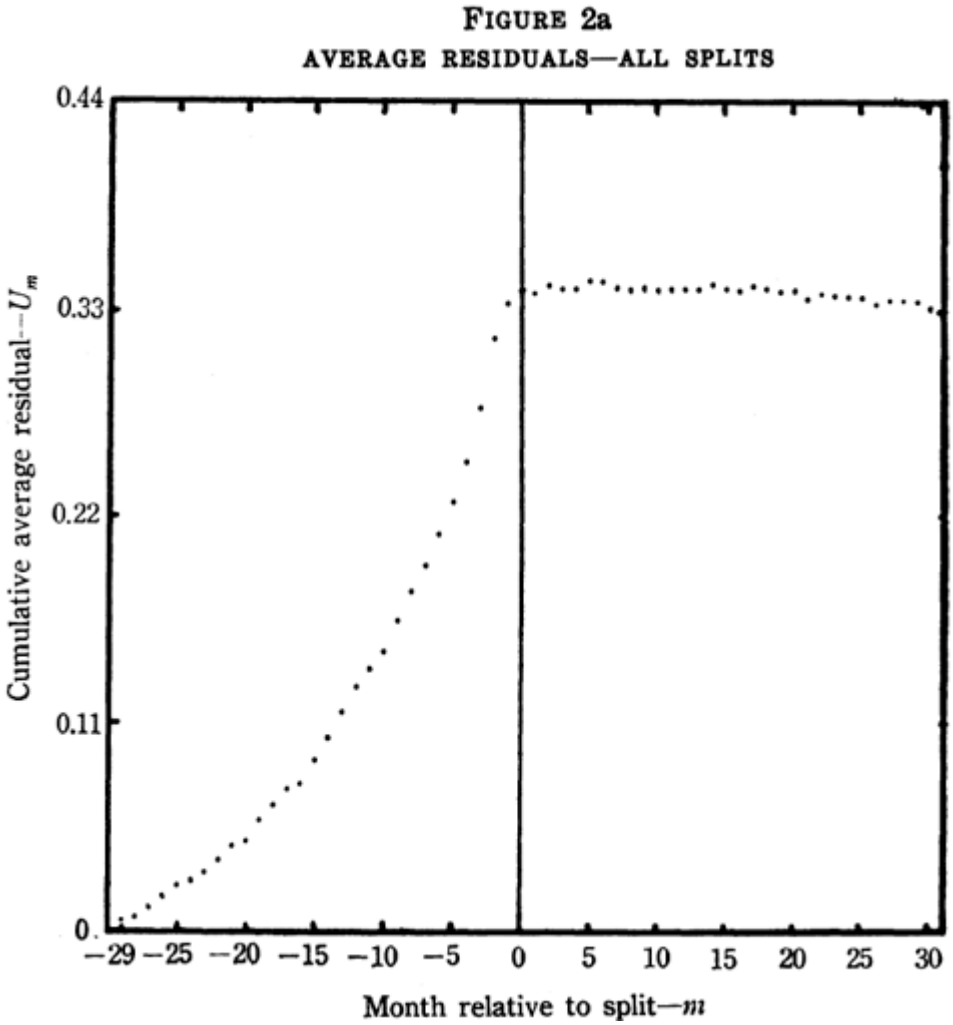
4.2 Introduction to Event Studies

- **What are Event Studies?**
  - A methodology to **isolate** and **quantify** the impact of a specific event (e.g., earnings announcement, merger) on asset prices.
  - They compare **actual returns** during an event window with the **expected (normal) returns**.

- **Definition:**
  An event study quantifies the impact of a specific event on asset prices by comparing **actual returns** with **expected (normal) returns**.
- **Core Idea:**
  Under normal conditions, returns follow a predictable pattern. An event disrupts this pattern, causing **abnormal returns (AR)**.

4.3 Examples (1 of 2)

- **Prominent event study:**
  Fama et al. (1969): Investigate the price adjustment following a stock split
  - Stock splits are preceded by abnormally high stock returns
  - After the stock split no abnormally high/low returns are observable
  - "The evidence indicates that on average the market's judgments concerning the information implications of a split are fully reflected in the price of a share [...]"



4.4 Examples (2 of 2)

- MacKinlay (1997): Cumulative abnormal returns around earnings anouncements.

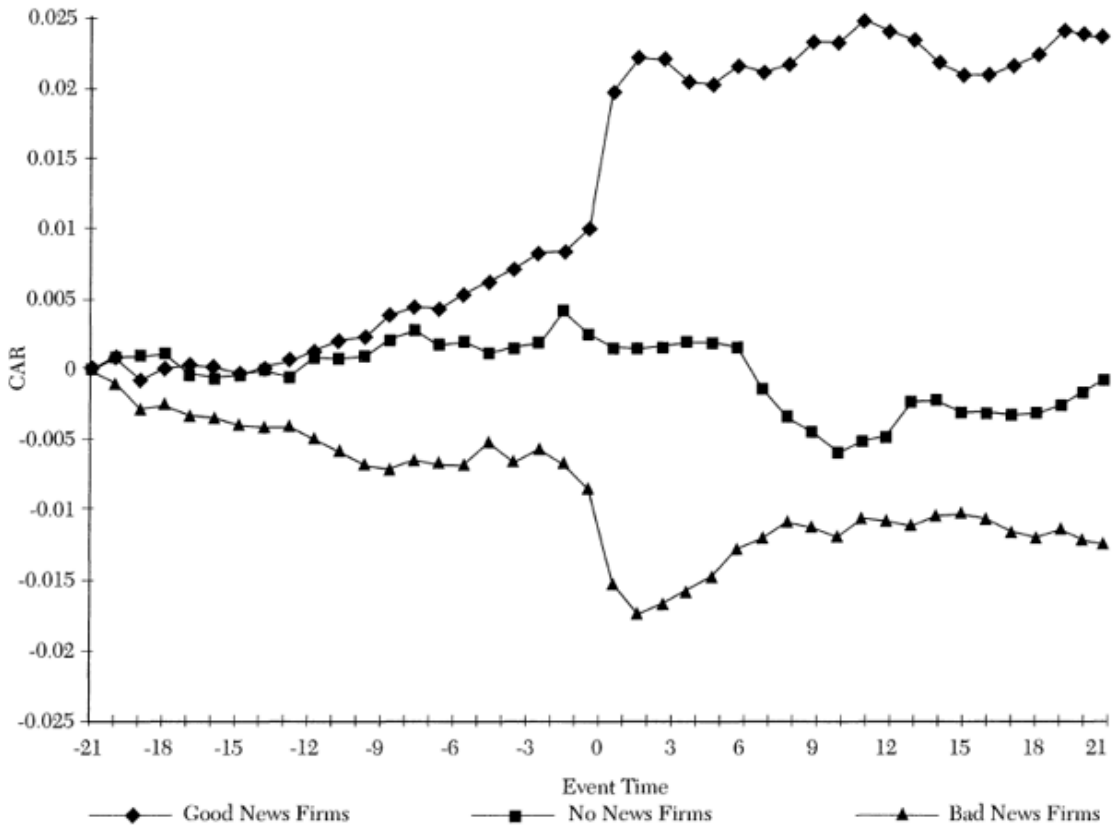


Figure 2a. Plot of cumulative abnormal return for earning announcements from event day -20 to event day 20. The abnormal return is calculated using the market model as the normal return measure.

Event studies allow us to evaluate whether and how quickly new information is incorporated into asset prices. They provide a systematic framework to test market efficiency by measuring abnormal returns.

4.5 Why Conduct Event Studies?

- **Objectives:**
  - Assess the speed and magnitude of price adjustment.
  - Determine if markets react **swiftly** and **accurately** to new information.
- **Key Questions:**
  - Are abnormal returns statistically significant?
  - How long does the impact of an event persist?

The main goal is to test whether prices adjust instantaneously and correctly to public information, a cornerstone of semi-strong market efficiency.

4.6 Historical Background

- **Early Work:**
  - Ball & Brown (1968)**:** Pioneered the use of event studies to evaluate earnings announcements.
  - Fama et al. (1969)**:** Extended these methods.
- **Evolution:**
  - Transition from simple announcement studies to multifactor models.

These early studies established the methodology that has evolved into today's sophisticated event studies, incorporating different asset pricing models and rigorous statistical testing.

4.7 Overview of the Event Study Framework

1. **Estimate Normal Returns:**
   Use an estimation window to model the "normal" behavior of returns.
2. **Calculate Abnormal Returns:**

$$AR_t = R_t - E(R_t)$$

3. **Aggregate Abnormal Returns:**

   - **Cumulative Abnormal Return (CAR):**

$$CAR_{[t_1,t_2]} = \sum_{t=t_1}^{t_2} AR_t$$

   - **Cumulative Average Abnormal Return (CAAR):**

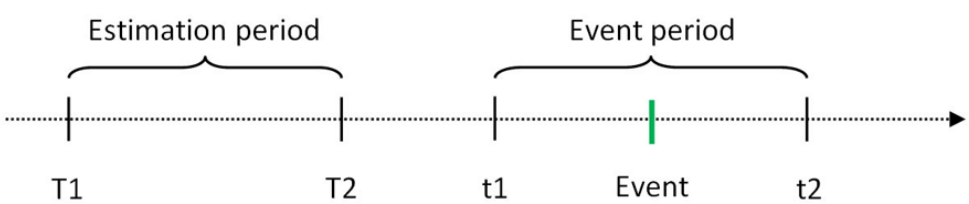
$$CAAR_{[t_1,t_2]} = \frac{1}{N}\sum_{i=1}^{N} CAR_{i,[t_1,t_2]}$$

4. **Statistical Testing:**
   Test whether AR, CAR, and CAAR differ significantly from zero.

This slide outlines the basic steps. The key challenge is estimating $E(R_t)$ accurately, so that the computed AR_t isolates the event's impact.

4.8 Defining the Estimation and Event Windows

- **Estimation Window:**
  - Period used to estimate the parameters of the chosen normal return model.
  - Typically extends prior to the event window to avoid contamination.
- **Pre-Event Window:**
  - A short period immediately before the event.
  - Used to check for any anticipatory effects or leakage of information.
- **Event Window:**
  - The period during which the event is expected to affect returns.
  - Must be long enough to capture the full reaction but short enough to minimize noise.
- **Post-Event Window:**
  - The period following the event.
  - Assesses whether the impact persists, reverses, or stabilizes.



The careful definition of windows is crucial. The estimation window must be free from the event's influence to provide a reliable baseline. The pre-event window checks for potential information leakage, while the event and post-event windows capture the market's reaction and any lingering effects.

4.9 Estimating Normal Returns – Introduction

- **Goal:**
  Determine what returns would have been in the absence of the event.

- **Basic Formula:**

$$AR_t = R_t - E(R_t)$$

- **Challenge:**
  The quality of AR_t depends on the model used for $E(R_t)$.

> Before the event, we assume that returns follow a known process. Our task is to model this "normal" behavior to later compare it with actual returns.

4.10 Estimating Normal Returns – Model Options (Part 1)

Mean-Adjusted Model:

- **Assumption:** Returns are constant on average.
  - **Formula:** $E(R_t) = \bar{R}$
- **Pros & Cons:**
  - **Pros:** Simple to implement.
  - **Cons:** Ignores time-varying risk factors.

Market Model:

- **Assumption:** A linear relation exists between asset returns and market returns.
  - **Formula:** $R_t = \alpha + \beta R_{mt} + \epsilon_t$
  - **Estimation:** Parameters α and β are estimated via Ordinary Least Squares (OLS) over the estimation window.

> The mean-adjusted model is a benchmark; however, the market model is more widely used because it accounts for the systematic risk component via R_{mt}.

4.11 Estimating Normal Returns – Model Options (Part 2)

Multifactor Models:

- **Examples:**
  - Fama-French 3-Factor Model:

$$R_t = \alpha + \beta_1 R_{mt} + \beta_2 SMB_t + \beta_3 HML_t + \epsilon_t$$

- Carhart 4-Factor Model (adds momentum):

$$R_t = \alpha + \beta_1 R_{mt} + \beta_2 SMB_t + \beta_3 HML_t + \beta_4 MOM_t + \epsilon_t$$

- **Advantage:** Better risk adjustment by incorporating multiple risk factors.
- **Consideration:** Increased complexity and potential for overfitting.

Multifactor models allow a more nuanced estimation of $E(R_t)$. By including additional factors, these models capture more of the systematic risk affecting returns.

4.12 Considerations in Model Selection

- **Goodness-of-Fit:**
  Evaluate the R^2 and residual properties to ensure the model explains most of the return variation.
- **Estimation Window:**
  Choose a window that is long enough for precision but avoids structural breaks.
- **Robustness:**
  Check whether the abnormal returns are consistent across different model specifications.
- **Trade-Off:**
  Simplicity versus complexity—balance parameter parsimony and accuracy.

The choice of model is crucial. An improper model can misattribute normal variations to the event, thereby distorting AR estimates. Robustness checks across different models increase confidence in the findings.

4.13 Types of Event Studies

- **Common Event Date Studies:**
  All firms experience the event on the same calendar day (e.g., macroeconomic announcements).
- **Staggered Event Studies:**
  Firm-specific events occur on different dates (e.g., earnings announcements, M&A deals).
- **Panel Event Studies vs. Individual Studies:**
  - **Panel:** Aggregate data across multiple firms.
  - **Individual:** Analyze each firm separately and then summarize.

The design depends on the nature of the event. Staggered events require aligning each firm's data relative to its event date, while common event date studies are simpler to aggregate.

4.14 Testing Significance of AAR and CAAR

Vector Testing

- **Average Abnormal Return (AAR):** $AAR_t = \frac{1}{N}\sum_{i=1}^{N} AR_{i,t}$
- **Cumulative Average Abnormal Return (CAAR):** $CAAR_{[t_1,t_2]} = \frac{1}{N}\sum_{i=1}^{N}\sum_{t=t_1}^{t_2} AR_{i,t}$
- **Testing Procedure:**
  - Compute the standard error of AAR/CAAR.
  - **t-Statistic:**
  $$t = \frac{AAR_t}{SE(AAR_t)} \quad \text{or} \quad t = \frac{CAAR}{SE(CAAR)}$$
  - **Null Hypothesis:**
  $$H_0 : AAR_t = 0 \quad \text{or} \quad CAAR_{[t_1,t_2]} = 0$$

Once we have a vector of ARs across events, we average them to form AAR and then aggregate to CAAR. Under the null hypothesis, both AAR and CAAR should be zero. The t-statistic is used for hypothesis testing, assuming that ARs are approximately normally distributed.

4.15 Plotting AAR and CAAR

- **Visualization:**
  - **Line Plot of AAR:** Shows daily average abnormal returns.
  - **Line Plot of CAAR:** Shows the accumulation of abnormal returns over time.
- **Confidence Intervals:**
  - Use standard errors or bootstrapped intervals.
- **Interpretation:**
  - Significant deviations from zero indicate an event effect.

Graphical displays help in visualizing the dynamic effects of an event. Confidence intervals around the AAR/CAAR plots enable us to assess statistical significance visually.

4.16 Example (MacKinlay, 1997)

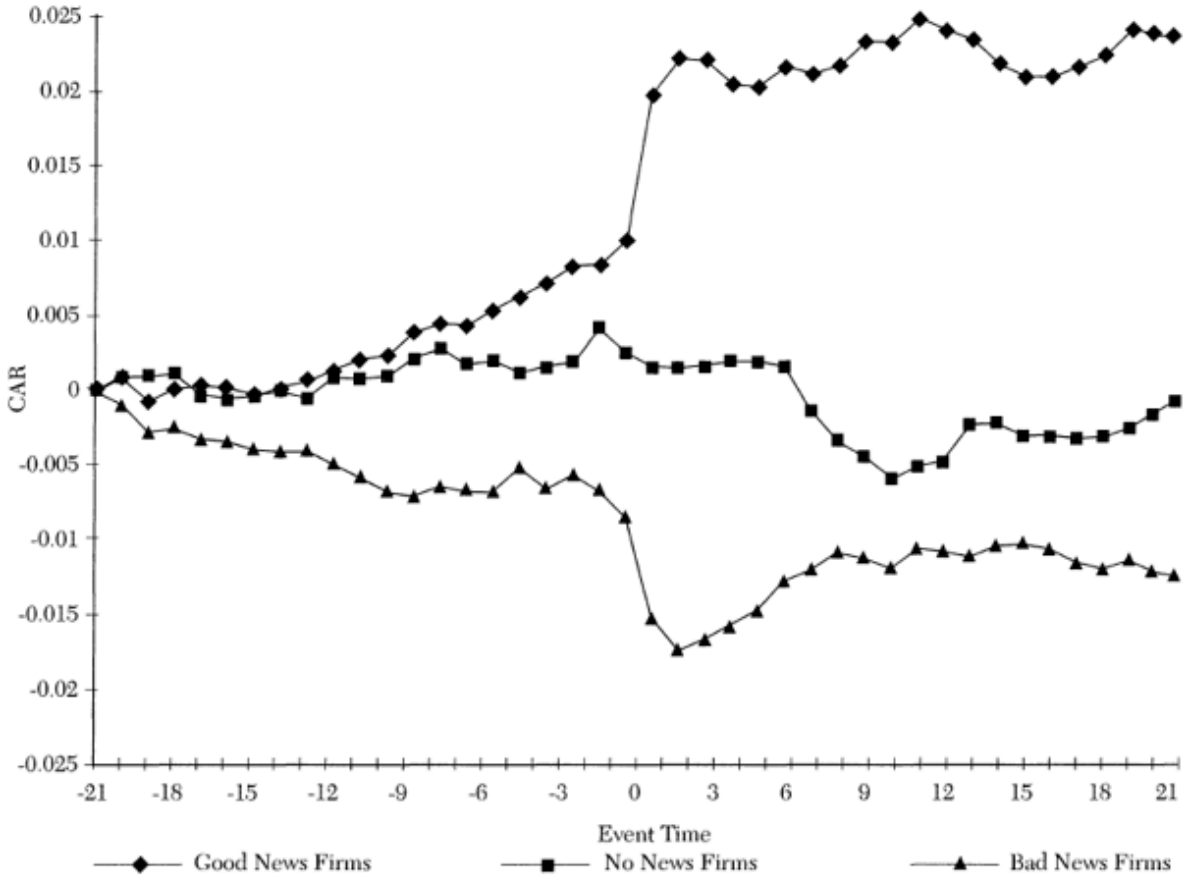


Figure 2a. Plot of cumulative abnormal return for earning announcements from event day -20 to event day 20. The abnormal return is calculated using the market model as the normal return measure.

4.17 R Code – Calculating AR, CAR, and CAAR

```
# Load required packages
library(dplyr)
library(ggplot2)

# Set event date and windows (example dates)
event_date <- as.Date("2020-11-03")
estimation_start <- event_date - 180
estimation_end <- event_date - 60
event_start <- event_date - 10
```

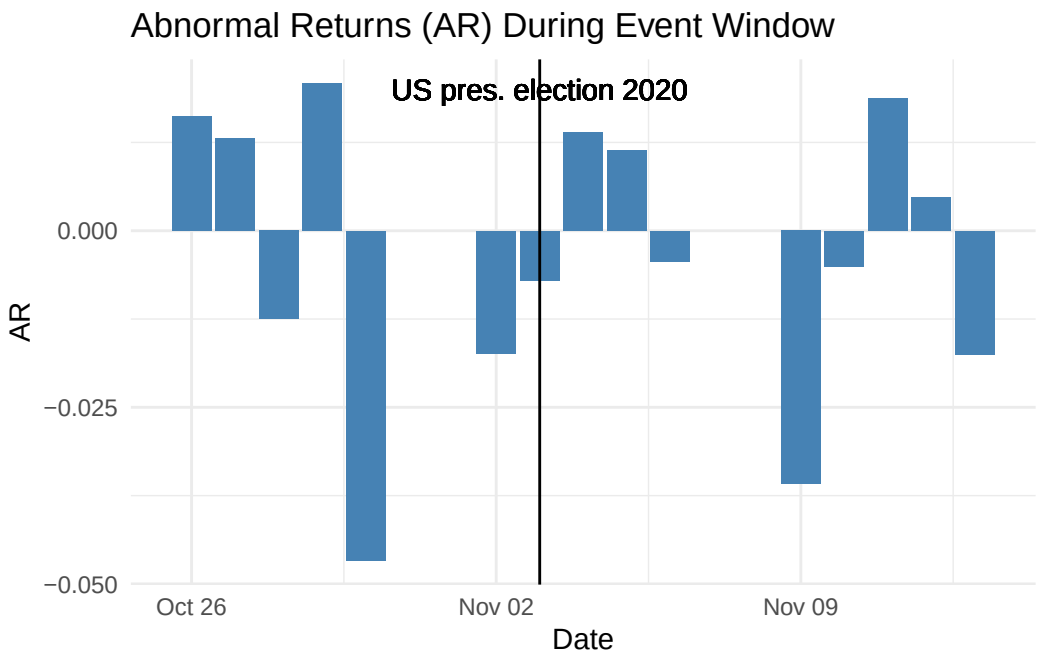
```r
event_end <- event_date + 10

# Download AAPL and SPY data
load("stocks.RData")

# Calculate daily returns
aapl <- stocks_data %>%
  filter(symbol == "AAPL") %>%
  arrange(date) %>%
  mutate(R = adjusted / lag(adjusted) - 1) |>
  select(symbol, date, R)
spy <- stocks_data %>%
  filter(symbol == "^GSPC") %>%
  arrange(date) %>%
  mutate(Rm = adjusted / lag(adjusted) - 1) |>
  select(date, Rm)

# Merge data
data <- inner_join(aapl, spy, by = "date")

# Classify periods
data <- data %>% mutate(period = case_when(
  date >= estimation_start & date <= estimation_end ~ "Estimation",
  date >= event_start & date <= event_end ~ "Event",
  TRUE ~ "Other"
))

# Estimate market model using estimation window
est_data <- data %>%
  filter(period == "Estimation") %>%
  drop_na()
market_model <- lm(R ~ Rm, data = est_data)

# Compute expected returns and abnormal returns in event window
event_data <- data %>%
  filter(period == "Event") %>%
  drop_na()
event_data <- event_data %>%
  mutate(
    E_R = predict(market_model, newdata = event_data),
    AR = R - E_R
  )
```

```r
# Compute CAR over the event window
CAR <- sum(event_data$AR)
# (For CAAR, with multiple events, average CARs across events)
cat("Cumulative Abnormal Return (CAR):", round(CAR, 4), "\n")

# t-test AAR before vs after event
t_test <- t.test(
  event_data |> filter(date < event_date) |> pull(AR),
  event_data |> filter(date >= event_date) |> pull(AR)
)
cat("t-Test for AAR before vs. after event:\n")
print(t_test)
```

```
Cumulative Abnormal Return (CAR): -0.0474
t-Test for AAR before vs. after event:

    Welch Two Sample t-test

data:  pull(filter(event_data, date < event_date), AR) and pull(filter(event_data, date >= ev
t = -0.16995, df = 7.8715, p-value = 0.8693
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -0.02996290  0.02586047
sample estimates:
   mean of x    mean of y
-0.004393452 -0.002342237
```

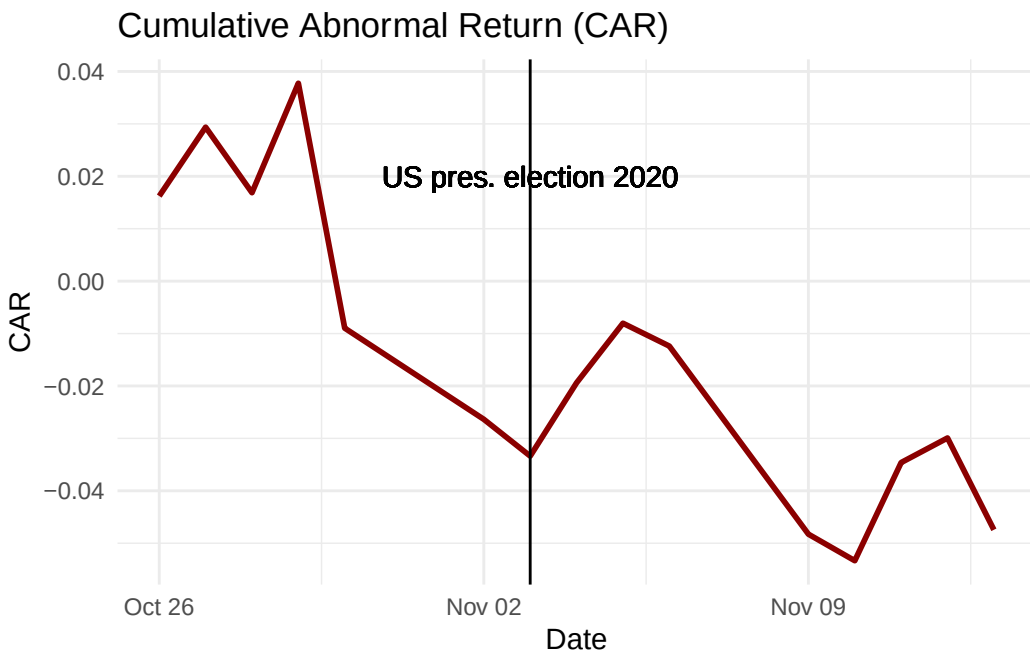
4.18 R Code - Plotting AR and CAR

```r
# Plot AR
ggplot(event_data, aes(x = date, y = AR)) +
  geom_bar(stat = "identity", fill = "steelblue") +
  labs(title = "Abnormal Returns (AR) During Event Window", x = "Date", y = "AR") +
  # add vertical black line with text "US pres. election 2020"
  geom_vline(xintercept = as.Date("2020-11-03"), color = "black") +
  geom_text(aes(
    x = as.Date("2020-11-03"), y = 0.02,
    label = "US pres. election 2020"
  ), color = "black") +
  theme_minimal()
```



4.19 R Code - Plotting AR and CAR (2 of 2)

```
# Plot CAR
event_data <- event_data %>%
  arrange(date) %>%
  mutate(CAR = cumsum(AR))
ggplot(event_data, aes(x = date, y = CAR)) +
  geom_line(color = "darkred", size = 1) +
  labs(title = "Cumulative Abnormal Return (CAR)", x = "Date", y = "CAR") +
  # add vertical black line with text "US pres. election 2020"
  geom_vline(xintercept = as.Date("2020-11-03"), color = "black") +
  geom_text(aes(
    x = as.Date("2020-11-03"), y = 0.02,
    label = "US pres. election 2020"
  ), color = "black") +
  theme_minimal()
```



4.20 Testing for Event-Induced Volatility

- **Concept:**
  Events may also increase the variance of returns.
- **Procedure:**
  - Compare the variance during the event window ($sigma^2_{event}$) with the variance during the estimation window (σ^2_{normal}).
- **F-Test for Variance Equality:**

$$F = \frac{\sigma^2_{event}}{\sigma^2_{normal}}$$

- **Null Hypothesis:**

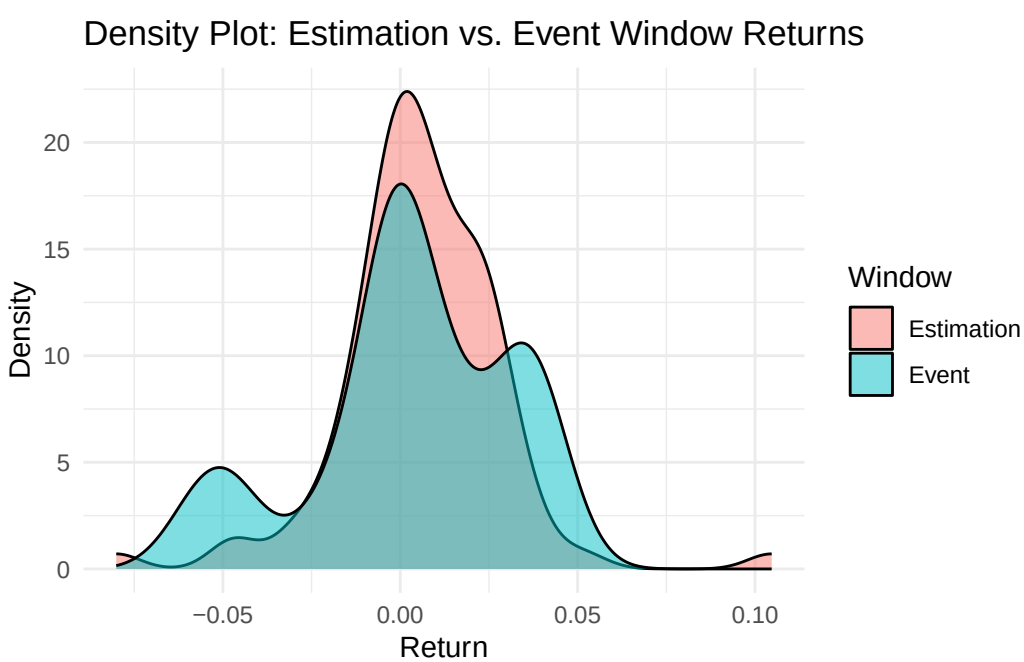
$$H_0 : \sigma^2_{event} = \sigma^2_{normal}$$

- **Corrections:**
  - Adjust for heteroskedasticity using robust standard errors.

An F-test compares the variance during the event window with that of the estimation window. If the event induces extra volatility, the F-statistic will be significantly different from 1. However, caution is needed when returns are not homoscedastic.

4.21 R Code - testing for Event-Induced Volatility

```
# Calculate variances for estimation and event windows
var_est <- var(est_data$R, na.rm = TRUE)
var_event <- var(event_data$R, na.rm = TRUE)

# Perform F-test for equal variances
f_test <- var.test(event_data$R, est_data$R)
print(f_test)

# Density plot for visual comparison
df_est <- data.frame(Returns = est_data$R, Window = "Estimation")
df_event <- data.frame(Returns = event_data$R, Window = "Event")
df_all <- rbind(df_est, df_event)

ggplot(df_all, aes(x = Returns, fill = Window)) +
  geom_density(alpha = 0.5) +
  labs(
    title = "Density Plot: Estimation vs. Event Window Returns",
    x = "Return", y = "Density"
  ) +
  theme_minimal()
```



```
    F test to compare two variances

data:  event_data$R and est_data$R
F = 1.5378, num df = 14, denom df = 84, p-value = 0.2309
alternative hypothesis: true ratio of variances is not equal to 1
95 percent confidence interval:
 0.7589008 3.9657552
sample estimates:
ratio of variances
          1.537785
```

The F-test evaluates whether volatility (variance) during the event window is significantly different from that in the estimation window. A significant result suggests event-induced volatility.

4.22 Testing for Cross-Correlation

- **Objective:**
  To determine if abnormal returns are correlated across firms.
- **Method:**
  Compute the cross-correlation function (CCF) between AR vectors of different stocks.
- **Test:**
  - Use a modified t-test to check if the cross-correlation is statistically different from zero.
- **Corrections:**
  - Adjust for multiple testing and serial correlation.

Cross-correlation tests reveal whether an event in one firm spills over to related firms. This analysis helps in identifying contagion or sector-wide effects.

4.23 Available Statistical Tests and Corrections

- **t-Tests:**
  For AR, CAR, and CAAR with normality assumptions.
- **Nonparametric Tests:**
  E.g., Wilcoxon signed-rank test, useful when normality is in doubt.

- **Bootstrapping:**
  Generates empirical confidence intervals.
- **Corrections for Heteroskedasticity:**
  Use Newey-West or HAC (Heteroskedasticity and Autocorrelation Consistent) standard errors.
- **Multiple Testing Correction:**
  Bonferroni or Benjamini-Hochberg methods when testing multiple lags or multiple firms.

> No single test is perfect. Researchers must choose tests that are appropriate for the data characteristics and correct for issues such as heteroskedasticity and serial correlation. Bootstrapping provides a flexible, distribution-free alternative.

4.24 R Code Example

```r
# t-test for AR in the event window:
t_test_AR <- t.test(event_data$AR)
print(t_test_AR)

# Wilcoxon signed-rank test:
wilcox_test <- wilcox.test(event_data$AR)
print(wilcox_test)

# Bootstrapping for CAR confidence interval:
library(boot)
CAR_func <- function(data, indices) {
  d <- data[indices]
  return(sum(d))
}
boot_result <- boot(event_data$AR, CAR_func, R = 1000)
print(boot_result)
boot_ci <- boot.ci(boot_result, type = "perc")
print(boot_ci)

# Newey-West robust standard errors for the market model:
library(sandwich)
library(lmtest)
nw_se <- sqrt(diag(NeweyWest(market_model)))

print(summary(nw_se))
```

```
    One Sample t-test

data:  event_data$AR
t = -0.60453, df = 14, p-value = 0.5552
alternative hypothesis: true mean is not equal to 0
95 percent confidence interval:
 -0.014383611  0.008058165
sample estimates:
   mean of x
-0.003162723


    Wilcoxon signed rank exact test

data:  event_data$AR
V = 56, p-value = 0.8469
alternative hypothesis: true location is not equal to 0


ORDINARY NONPARAMETRIC BOOTSTRAP


Call:
boot(data = event_data$AR, statistic = CAR_func, R = 1000)


Bootstrap Statistics :
       original       bias    std. error
t1* -0.04744084 0.0002248712  0.07525067
BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
Based on 1000 bootstrap replicates

CALL :
boot.ci(boot.out = boot_result, type = "perc")

Intervals :
Level     Percentile
95%   (-0.1968,  0.0934 )
Calculations and Intervals on Original Scale
    Min.  1st Qu.   Median     Mean  3rd Qu.     Max.
0.002262 0.040034 0.077805 0.077805 0.115576 0.153347
```

This slide illustrates various statistical techniques. The t-test and Wilcoxon test assess the significance of abnormal returns, bootstrapping creates robust confidence intervals for CAR, and the Newey-West method provides robust standard errors in the presence of heteroskedasticity and autocorrelation. Adjustments for multiple testing control the false discovery rate.

4.25 Summary of the Event Study Methodology

- **Steps Recap:**
  1. **Estimate Normal Returns:** Choose a model (mean-adjusted, market, or multi-factor).
  2. **Define Windows:** Estimation, pre-event, event, and post-event.
  3. **Calculate AR, CAR, CAAR.**
  4. **Test for Significance:** t-tests, bootstrapping, nonparametric methods.
  5. **Examine Secondary Effects:** Event-induced volatility and cross-correlations.
- **Model and Window Sensitivity:**
  Robustness checks are essential.

This summary ties together all the components of an event study. The reliability of conclusions depends on careful model specification, window selection, and appropriate statistical testing.

4.26 Final Remarks and Practical Implications

- **Key Takeaways:**
  - Event studies are a powerful tool to assess market reactions.
  - Mathematical rigor in estimating normal returns and testing abnormal returns is essential.
  - Consideration of volatility and cross-sectional correlations deepens the analysis.
- **Practical Implication:**
  - These methods inform both academic research and investment decisions by providing evidence on market efficiency.
- **Ongoing Developments:**
  - Continuous improvements in econometric methods enhance the reliability of event studies.

Understanding the full methodology—from estimating normal returns to testing for abnormal volatility and cross-correlation—provides a deep insight into market behavior. This chapter equips you with the tools necessary to critically evaluate how markets process new information.

4.27 Final Takeaways

- **Mathematical Detail:**
  Each step in the event study framework is supported by precise mathematical formulations.
- **Empirical Rigor:**
  The combination of various tests and corrections ensures robust conclusions.
- **Application:**
  Event studies remain central in assessing market efficiency and the impact of information shocks.

4.28 References

Bachelier, L. (1900). *Théorie de la spéculation*.

Ball, R., & Brown, P. (1968). An Empirical Evaluation of Accounting Income Numbers. *Journal of Accounting Research*, *6*(2), 159–178. https://doi.org/10.2307/2490232

Black, F., & Scholes, M. (1973). The Pricing of Options and Corporate Liabilities. *Journal of Political Economy*, *81*(3), 637–654.

Fama, E. F. (1970). Efficient Capital Markets: A Review of Theory and Empirical Work. *The Journal of Finance*, *25*(2), 383–417. https://doi.org/10.2307/2325486

Fama, E. F., Fisher, L., Jensen, M. C., & Roll, R. (1969). The Adjustment of Stock Prices to New Information. *International Economic Review*, *10*(1), 1–21. https://doi.org/10.2307/2525569

Grossman, S. J., & Stiglitz, J. E. (1980). On the Impossibility of Informationally Efficient Markets. *The American Economic Review*, *70*(3), 393–408. https://www.jstor.org/stable/1805228

Lo, A. W., & MacKinlay, A. C. (1988). Stock Market Prices Do Not Follow Random Walks: Evidence from a Simple Specification Test. *Review of Financial Studies*, *1*(1), 41–66. https://doi.org/10.1093/rfs/1.1.41

MacKinlay, A. C. (1997). Event Studies in Economics and Finance. *Journal of Economic Literature*, *35*(1), 13–39. https://www.jstor.org/stable/2729691

Malkiel, B. G. (1989). Efficient Market Hypothesis. In J. Eatwell, M. Milgate, & P. Newman (Eds.), *Finance* (pp. 127–134). Palgrave Macmillan UK. https://doi.org/10.1007/978-1-349-20213-3_13

Roberts, H. (1967). *Statistical versus Clinical Prediction of the Stock Market – ScienceOpen* [Unpublished {{Manuscript}}].

Samuelson, P. A. (1965). Proof that properly anticipated prices fluctuate randomly. *Industrial Management Review*, *6*(2), 41–49.